

MySQL 8.0 Replication Compatibility Review

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Scope: Glial + PmaControl replication collection and legacy master/slave compatibility

Goal

Make MySQL 8.0 replica metadata usable everywhere in PmaControl even when the server exposes:

- `Source\_Host`
- `Source\_Port`
- `Replica\_IO\_Running`
- `Replica\_SQL\_Running`

instead of the legacy fields:

- `Master\_Host`
- `Master\_Port`
- `Slave\_IO\_Running`
- `Slave\_SQL\_Running`

The immediate functional goal was to add `source\_host` and `source\_port` compatibility without breaking the older code that still expects `master\_host` and `master\_port`.

Code Review Findings

P1. Glial returned raw `SHOW REPLICA STATUS` rows without legacy aliases

Files reviewed:

- /srv/www/glial/Glial/Sgbd/Sql/Mysql/Mysql.php
- /srv/www/glial/Glial/Sgbd/Sql/Mysql/MasterSlave.php

Problem:

- Glial already switches to `SHOW REPLICA STATUS` on newer MySQL 8.0 servers.
- The returned array was then consumed by older code paths that still read keys such as `Master\_Host`, `Master\_Port`, `Slave\_IO\_Running`, `Relay\_Master\_Log\_File`, `Exec\_Master\_Log\_Pos`, and `Connection\_name`.
- That created a compatibility gap: transport was modern, consumers were still legacy.

Impact:

- replication topology code can silently lose source endpoint information
- failover tooling can read empty host/port values
- operational CLI code can misinterpret replication state

P1. PmaControl integration stored only legacy master/slave names

File reviewed:

- /srv/www/pmacontrol/App/Controller/Integrate.php

Problem:

- the integration layer lowercases replication rows and normalizes `connection\_name`
- but before the fix it only guaranteed `master\_host` and `master\_port`
- `source\_host` and `source\_port` were not backfilled

Impact:

- newer MySQL 8.0 terminology was not preserved in the integrated slave metrics
- downstream code could not query `slave::source\_host` and `slave::source\_port`

P2. Several operational paths still emit legacy replication SQL

Files reviewed:

- /srv/www/pmacontrol/App/Controller/Load.php
- /srv/www/pmacontrol/App/Controller/Load2.php
- /srv/www/pmacontrol/App/Controller/Mysql.php
- /srv/www/pmacontrol/App/Controller/Slave.php
- /srv/www/glial/Glial/Neuron/PmaCli/PmaCliSwitch.php
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- /srv/www/glial/Glial/Neuron/PmaCli/PmaCliFailOver.php

Problem:

- these paths still use `SHOW SLAVE STATUS`, `STOP SLAVE`, `START SLAVE`, and `CHANGE MASTER TO`
- on current MySQL 8 they are mostly still accepted as compatibility syntax, but they are legacy terminology

Impact:

- no immediate break on most environments
- but this remains technical debt and should be migrated deliberately later

Implemented Fix

1. Glial replication row normalization

Added compatibility aliases in:

- /srv/www/glial/Glial/Sgbd/Sql/Mysql/Mysql.php
- /srv/www/glial/Glial/Sgbd/Sql/Mysql/MasterSlave.php

Normalization now mirrors both directions for the key fields:

- `Source\_Host` <-> `Master\_Host`
- `Source\_Port` <-> `Master\_Port`
- `Source\_Log\_File` <-> `Master\_Log\_File`
- `Read\_Source\_Log\_Pos` <-> `Read\_Master\_Log\_Pos`
- `Relay\_Source\_Log\_File` <-> `Relay\_Master\_Log\_File`
- `Exec\_Source\_Log\_Pos` <-> `Exec\_Master\_Log\_Pos`
- `Replica\_IO\_Running` <-> `Slave\_IO\_Running`
- `Replica\_SQL\_Running` <-> `Slave\_SQL\_Running`
- `Replica\_IO\_State` <-> `Slave\_IO\_State`
- `Replica\_SQL\_Running\_State` <-> `Slave\_SQL\_Running\_State`

- `Seconds\_Behind\_Source` <-> `Seconds\_Behind\_Master`
- `Channel\_Name` <-> `Connection\_name`

Result:

- old callers keep working
- new callers can read source/replica terminology
- mixed environments remain compatible

2. Integration layer normalization

Updated:

- /srv/www/pmacontrol/App/Controller/Integrate.php

Behavior now:

- if only `source\_host/source\_port` exist, `master\_host/master\_port` are populated
- if only `master\_host/master\_port` exist, `source\_host/source\_port` are populated
- `seconds\_behind\_source` and `seconds\_behind\_master` are kept in sync

Result:

- PmaControl can expose both naming conventions in extracted replication metrics

Tests Added

Updated/added:

- /srv/www/pmacontrol/tests/Controller/IntegrateTest.php
- /srv/www/pmacontrol/tests/Library/MysqlReplicationCompatibilityTest.php

Covered cases:

- MySQL 8.0 `SHOW REPLICA STATUS` row with `Source\_\*` and `Replica\_\*`
- backfill of legacy `Master\_\*` and `Slave\_\*`
- backfill of modern `source\_\*`
- `seconds\_behind\_source` and `seconds\_behind\_master`
- `Channel\_Name` / `Connection\_name`

Validation:

- `./vendor/bin/phpunit tests/Controller/IntegrateTest.php tests/Library/MysqlReplicationCompatibilityTest.php`
- result: `OK (6 tests, 33 assertions)`

Practical Outcome

After this change:

- Glial can read MySQL 8.0 replica rows and still feed legacy code safely
- PmaControl stores `source\_host` and `source\_port` compatibility fields
- topology, extraction, and replication displays no longer depend on one naming convention only

Residual Work

Not changed in this patch:

- command emitters still use `SHOW SLAVE STATUS`
- replication control SQL still uses `STOP SLAVE`, `START SLAVE`, `CHANGE MASTER TO`

Recommended next step:

1. introduce a dedicated compatibility wrapper for replication control commands
2. migrate write-side operations to source/replica terminology where supported
3. keep aliasing on the read path for long-term backward compatibility

Recommendation

Keep the alias layer in Glial permanently.

Reason:

- it centralizes compatibility once
- it avoids duplicated fixes in controllers
- it protects old operational tooling while allowing PmaControl to expose modern MySQL 8.0 field names